Chapter 14, Problem 69P

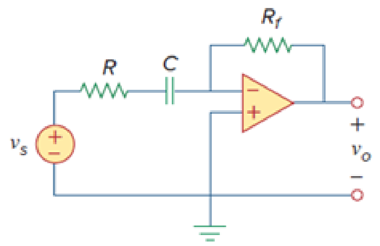
Problem

Design the filter in Fig. 14.94 to meet the following requirements:

(a) It must attenuate a signal at 2 kHz by 3 dB compared with its value at 10 MHz.

(b) It must provide a steady-state output of *vo*(*t*) = 10 sin(2*π* × 108*t* + 180°) V for an input *vs*(*t*) = 4 sin(2*π* × 108*t*) V.

Figure 14.94:



Step-by-step solution

Step 1 of 7

Refer to Figure 14.94 in the textbook.

As per requirement (a), it attenuates low frequencies and allows high frequencies. So, the circuit is High pass filter.

The input impedance of the circuit is,

$$Z_i = R + \frac{1}{j\omega C}$$

The feedback impedance of the circuit is,

$$Z_f = R_f$$

Step 2 of 7

Calculate the expression for the gain of the circuit.

$$\begin{aligned}\frac{V_o}{V_i} &= -\frac{Z_f}{Z_i} \\ &= -\frac{R_f}{R + \frac{1}{j\omega C}} \\ &= -\frac{\frac{R_f}{R}}{1 + \frac{1}{j\omega RC}} \\ &= -\frac{\frac{R_f}{R}}{1 - j\left(\frac{\omega_c}{\omega}\right)}\end{aligned}$$

Step 3 of 7

The input voltage of the circuit is,

$$v_s(t) = 4\sin\left(2\pi \times 10^8 t\right) \text{ V}$$

The phasor value of input voltage is,

$$\begin{aligned} V_s &= 4\angle 0^\circ \text{ V} \\ &= 4 \text{ V} \end{aligned}$$

The output voltage of the circuit is,

$$v_o(t) = 10\sin\left(2\pi \times 10^8 t + 180^\circ\right) \text{ V}$$

The phasor value of output voltage is,

$$\begin{aligned} V_o &= 10\angle 180^\circ \text{ V} \\ &= -10 \text{ V} \end{aligned}$$

Step 4 of 7

The gain of the circuit is,

$$\begin{aligned} \frac{V_o}{V_s} &= \frac{-10}{4} \\ &= -2.5 \text{ V/V} \end{aligned}$$

The cutoff frequency of the filter is,

$$f_c = 2 \times 10^3$$
$$2\pi f_c = 4\pi \times 10^3$$
$$\omega_c = 4\pi \times 10^3$$
$$\frac{1}{RC} = 4\pi \times 10^3$$

Step 5 of 7

The gain of the circuit at high frequency, $\omega = 10 \text{ MHz}$ is,

$$-\left.\frac{\frac{R_f}{R}}{1 - j\left(\frac{\omega_c}{\omega}\right)}\right|_{\omega=10 \text{ MHz}} = -2.5$$

$$-\left.\frac{\frac{R_f}{R}}{1 - j\left(\frac{2\pi f_c}{\omega}\right)}\right|_{\omega=10 \text{ MHz}} = -2.5$$

$$-\left.\frac{\frac{R_f}{R}}{1 - j\left(\frac{2\pi\left(2 \times 10^3\right)}{10 \times 10^6}\right)}\right|_{\omega=10 \text{ MHz}} = -2.5$$

At high frequencies, the imaginary term is very close to zero and can be neglected.

$$-\frac{R_f}{R} = -2.5$$

$$R = \frac{R_f}{2.5}$$

Assume the feedback resistor as $R_f = 25 \text{ kHz}$.

Step 6 of 7

Calculate the value of resistor, R.

$$
\begin{aligned}
R &= \frac{R_f}{2.5} \\
&= \frac{25\ \text{k}\Omega}{2.5} \\
&= 10\ \text{k}\Omega
\end{aligned}
$$

Thus, the value of resistors are $\boxed{R = 10\ \text{k}\Omega}$ and $\boxed{R_f = 25\ \text{kHz}}$.

Step 7 of 7

Calculate the value of the capacitor, C.

$$
\frac{1}{RC} = 4\pi \times 10^3
$$

$$
C = \frac{1}{\left(4\pi \times 10^3\right) R}
$$

Substitute $10\ \text{k}\Omega$ for R.

$$
\begin{aligned}
C &= \frac{1}{\left(4\pi \times 10^3\right)\left(10 \times 10^3\right)} \\
&= 7.96 \times 10^{-9} \\
&= 7.96\ \text{nF}
\end{aligned}
$$

Thus, the value of the capacitor is $\boxed{C = 7.96\ \text{nF}}$.